Computability

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1. Algorithms and Existence of Non-Computable Functions

- **Effective Procedure**: It's a step by step method that solves a problem or evaluates a mathematical function. An effective procedure has a set of four strict formal requirements:
  - **Finite Instruction Set**: Defined by a finite list of clear, explicit instructions.
  - **Deterministic Execution**: Each step can be executed deterministically by a machine without relying on guesswork or insight.
  - **Finite Steps**: The computation must *halt* in a finite number of steps.
  - **Correct Output**: When the procedure halts, the result must be correct and unambiguous.

- **Computable Function**: It's a mathematical function $f : \mathbb{N}^k \longrightarrow \mathbb{N}$ for which there exists an effective procedure that calculates the correct output for any input the function's domain.
  - **Examples of Computable Functions**:
    - $\mathrm{GCD}(x, y) = \text{Greatest Common Divisor}$ it's solvable by the Euclid's Algorithm.
    - $f(n) = \begin{cases} 1 & \text{n is prime} \\ 0 & \text{otherwise} \end{cases}$
    - $g(n) = n^{\text{th}}$ digit in π
    - $h(n) = \begin{cases} 1 & \text{if } \pi \text{ includes at least } n \text{ digits 5 in a row} \\ 0 & \text{otherwise} \end{cases}$
  - **Examples of Functions of Unknown Computability**:
    - $q(n) = \begin{cases} 1 & \text{if in } \pi \text{ there are exactly n consecutive digits 5} \\ 0 & \text{otherwise} \end{cases}$

- **Why is $h(n)$ Computable and $q(n)$ is unknown?**:
  - **How do we know that $h(n)$ is Computable?**

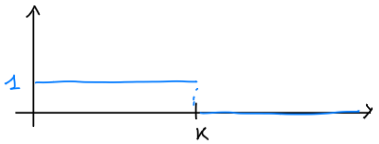
    Given the function $g(n) = \begin{cases} 1 & \text{if } \pi \text{ includes at least } n \text{ digits 5 in a row} \\ 0 & \text{otherwise} \end{cases}$

    This function has the following property: $g(n) = 1 \Longrightarrow \forall n' \leq n \quad g(n') = 1$.

    We can take $k = \sup\{n \mid \text{there are exactly } n \text{ digits 5 in a row in } \pi\}$ which gives us two possibilities:
    - k is finite

```
def g(n):
  if n <= k:
    return 1
  else:
    return 0
```

$$g(n) = \begin{cases} 1 & \text{if } n \leq k \\ 0 & \text{otherwise} \end{cases}$$



    - k is infinite

```
def g(n):
  return 1
```

$$g(n) = \begin{cases} 1 & \text{if } n \leq k \\ 0 & \text{otherwise} \end{cases}$$



    All of this tells us that a procedure exists, we don't know which one is it but it exists which is sufficient to say that $h(n)$ is computable.